

AKMAL ARIQ

Data & Analytics Engineer | Pipelines, Warehouses, Lakehouse & AI-Augmented Workflows

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83% regulatory report runtime cut · 100% OJK/BI submission accuracy · 20+ production pipelines · Zero-downtime orchestration migration · IELTS 7.5

PROFESSIONAL SUMMARY

Data & Analytics Engineer with 4 years of experience across banking and retail, delivering end-to-end data transformation — from profiling and mapping to quality assurance and reporting — that turns complex business requirements into reliable, trusted data assets. Design and operate pipelines integrating files, REST APIs, and RDBMS sources into a standardized enterprise view on both on-prem SQL Server and a modern lakehouse stack (S3-compatible object storage, ClickHouse, dbt, Dagster, Databricks). Production experience running 20+ pipelines across Airflow, Dagster, Talend, and SSIS; led the zero-downtime migration of 20+ production pipelines to Dagster + PostgreSQL. Achieved an 83% reduction in a key regulatory report's runtime and maintained 100% accuracy on OJK and Bank Indonesia submissions. Owned BRD development and solution design for ETL and Outsystems integrations, and established the team's first data engineering standards. Apply LLMs and retrieval-augmented generation (RAG) to data engineering — automating data-quality checks and building AI agents that validate data against documented business rules. Experienced collaborating with data governance, analytics, and business stakeholders. Fluent English (IELTS 7.5).

CORE COMPETENCIES

Pipelines & Orchestration: Airflow, Dagster, Kestra, SSIS, Talend, REST API, file- and event-based ingestion (JSON, CSV, XML), ETL/ELT design, scheduled job chaining

Data Modeling & Warehousing: Star Schema, SCD Type 2, partitioning, clustering, incremental loads (CDC, upsert, append, truncate), One Big Table

Data Quality & Governance: Data Profiling, Data Mapping, Validation & Reconciliation, OJK/BI regulatory reporting, PII compliance, dbt tests, Documentation, Master Data Alignment

AI & Agentic Engineering: LLM integration, RAG pipelines, semantic embeddings, vector retrieval (ChromaDB), AI agents for data-quality automation, prompt engineering, LLM-assisted coding

Collaboration & Leadership: BRD development, solution design, requirements gathering, stakeholder communication, mentoring, code reviews, Agile

Cloud & Data Platform: S3-compatible object storage (RustFS), ClickHouse, Databricks (Delta Live Tables, Unity Catalog, DAB), DuckDB, BigQuery & GCS, Medallion Architecture, OLAP Warehousing, Parquet/Iceberg

Languages & Automation: Python (Pandas, PyArrow, boto3, pyodbc, pymysql), Advanced SQL (T-SQL), Bash, GitHub Actions CI/CD, Jenkins, Docker, Linux/Unix, SSH, SFTP/SCP

Visualization & Reporting: Power BI, Tableau, Superset, Looker Studio, KPI dashboards, Regulatory Reporting (OJK/BI)

Observability: Grafana, Prometheus, structured logging, Telegram failure alerts, pipeline & server health monitoring

Marketplace & Ad Platforms: Shopee API, TikTok Shop API, Google Ads, Meta Ads, TikTok Ads

PROFESSIONAL EXPERIENCE

Kanmo Group | Jakarta, Indonesia

Senior Analytics Engineer | Mar 2026 – Present

- Engineered a data lakehouse pipeline across S3-compatible object storage, ClickHouse, dbt, and Dagster, standardizing marketplace sales, ad performance, seller, and live-stream data from Shopee and TikTok Shop into a trusted, queryable layer for analytics and BI.
- Scaffolding a modern OLAP layer on the lakehouse to power enterprise Power BI dashboards, enabling self-serve business metrics for CRM, marketing, and operations teams.
- Automated Power BI dashboards, providing real-time visibility into sales and customer engagement across retail brands.
- Built production API pipelines for Shopee (inventory snapshots, seller scores, automated token refresh) and TikTok Shop (live-stream analytics, daily inventory snapshots, historical backfill), plus daily Google/Meta/TikTok Ads pipelines feeding sales performance dashboards.
- Leading a proof-of-concept Databricks lakehouse using Delta Live Tables and Unity Catalog, evaluating open-source lakehouse architectures as a contingency.

Kanmo Group | Jakarta, Indonesia

Senior Data Engineer | Nov 2025 – Mar 2026

- Led the zero-downtime migration of legacy MySQL + Airflow orchestration to Dagster + PostgreSQL across 20+ production pipelines, owning end-to-end delivery and ensuring reliable, timely data for analytics.
- Architected a star-schema OLAP warehouse supporting CRM and marketplace dashboards, standardizing data structures across the organization.
- Established the team's first dedicated data engineering standards — pipeline structure, documentation, and coding standards — and instrumented pipelines with structured logging and Telegram failure alerts for 24/7 monitoring.
- Designed end-to-end Airflow pipelines consolidating files, REST APIs, and RDBMS sources from multiple retail brands into a unified Power BI semantic model for enterprise-wide reporting.
- Deployed a Grafana and Prometheus observability stack for real-time pipeline health monitoring, data freshness alerts, and infrastructure observability.
- Set up GitHub Actions runners for CI/CD, and managed Docker Compose stacks, PostgreSQL metadata stores, and backup procedures across Linux servers.

PT Bank SMBC Indonesia Tbk | Jakarta, Indonesia

Data Engineer | May 2023 – Nov 2025

- Owned BRD development and solution design for ETL pipelines and Outsystems web-app integrations within the Digital Service Enablement team, translating business requirements into deliverable technical solutions.
- Optimized on-prem SQL Server data models, cutting a key regulatory report's runtime by 83% (2 hours → 20 minutes) through schema refactoring, partitioning, and targeted indexing across 10+ models.
- Owned end-to-end reconciliation and data quality validation for OJK and Bank Indonesia reporting, maintaining 100% data accuracy and compliance in partnership with MIS and regulatory teams.
- Engineered the RPIM regulatory ETL pipeline end-to-end (SSIS, T-SQL) under strict PII / data-governance standards, and maintained 5+ batch integrations feeding OJK/BI reporting marts.
- Engineered a high-reliability Fund Transfer Processing batch ETL pipeline with scheduled job chaining, error-handling, and monitoring; production SLA consistently met.

- Migrated a production MySQL database to SQL Server using SSMA and backup/restore, moving core application data to the enterprise platform with minimal downtime.
- Mentored junior engineers through code reviews and pairing, raising team standards.

PT Bank Sinarmas Tbk | Jakarta, Indonesia

IT Graduate Camp (Data Engineer) | Aug 2022 – Mar 2023

- Built Talend and Airflow ETL workflows (Python: Pandas, NumPy) for the graduate camp, constructing a data warehouse from public marketplace data and delivering Tableau dashboards end to end.

PT Bank Mandiri (Persero) Tbk. | Jakarta, Indonesia

Internal Audit Intern | Mar 2021 – Sep 2021

- Tracked and summarized weekly outstanding reports for the Retail Audit Group in Excel, delivering clear, concise summaries for the team's reporting cycle.

Bandung Techno Park | Bandung, Indonesia

Data Scientist Intern | Jun 2020 – Jul 2020

- Performed exploratory data analysis on e-commerce data, building the analytical foundation used across later data engineering roles.

Telkom University | Bandung, Indonesia

Teaching & Lab Assistant | 2018 – 2021

- Teaching Assistant (Automata & Language Theory, 2021) and Computer Lab Assistant (Data Structures, 2018–2019), supporting course delivery and building early teaching and communication experience.

KEY PROJECTS

Data Lakehouse (RustFS → ClickHouse → dbt → Superset) | Kanmo, Production 2025–2026

- Built and ran a Bronze → Silver → Gold lakehouse pipeline orchestrated by Dagster, demonstrating object-storage ingestion, analytical transformation with dbt, and BI consumption end to end.

Agentic Data Quality Pipeline | GitHub, Public

2026

- Designed and built an agentic data engineering system: an LLM/RAG agent that ingests raw data, retrieves documented business rules from a vector index, detects data-quality issues, explains findings, and applies safe fixes — orchestrated with Dagster, going beyond traditional ETL.
- Implemented dual agent backends (offline heuristic + OpenAI-compatible LLM), a lightweight RAG index over business rules, automated issue detection, and safe auto-fixes; shipped with a full test suite.

Data Catalog RAG Agent | GitHub, Public · CI passing

2026

- Built an LLM/RAG data catalog assistant: a ChromaDB vector database stores the warehouse's data dictionary; an AI agent retrieves table/column definitions and answers analysts' questions with cited sources — orchestrated with Dagster.

Retail Demand Forecasting with AI | GitHub, Public · CI passing

2026

- Exploring store-level demand forecasting for multi-branch retail: gradient-boosted daily sales forecasting (walk-forward validation, MAE/RMSE/MAPE) with a GenAI layer that turns forecasts into store-specific operational recommendations — orchestrated with Dagster.

Self-Hosted AI Chat Platform | Personal Initiative

2026

- Built a production-grade AI chat application (FastAPI, MinIO, JWT auth, SSE streaming) with persistent memory, context compaction, and file analysis — Python code execution against attached data (CSV, Excel, PDF).

Earthquake Analytics Pipeline | DataTalks.Club Zoomcamp 2026

2026

- End-to-end GCP data platform on Terraform, Kestra, GCS, and BigQuery: medallion warehouse (partitioned/clustered), dbt transformations, and Looker Studio dashboards analyzing 500K+ seismic records — fully reproducible IaC.

FreshCart E-commerce Analytics Pipeline | Portfolio (akmalariq.dev)

2026

- Production-grade Airflow (Docker LocalExecutor) pipeline ingesting 150K+ records to GCS and BigQuery, with dbt Bronze→Silver→Gold modeling, SQL-based RFM segmentation, and 20+ automated data-quality assertions.

E-commerce Data Analysis | Portfolio

2026

- Explored and cleaned a public e-commerce dataset in SQL, built an RFM customer segmentation, and delivered a Power BI dashboard of sales trends.

Modern Data Warehouse with DuckDB | Personal Initiative

2026

- Built a Bronze → Silver → Gold pipeline using the Medallion Architecture, demonstrating end-to-end data modeling and analytical workflow.

Batch Data Pipeline | Personal Initiative

2025

- Airflow, DuckDB, Delta Lake, Trino, MinIO, and Metabase — medallion warehouse with full observability and data quality.

MySQL → SQL Server Migration | PT Bank SMBC Indonesia Tbk, Production

2023–2025

- Migrated a production MySQL database to SQL Server using SSMA and backup/restore, moving core application data to the enterprise platform with minimal downtime.

EDUCATION

Telkom University | Bandung, Indonesia — Bachelor of Engineering (B.E.) in Informatics, GPA 3.49/4.00 (2022)

CERTIFICATIONS & LANGUAGES

- **Databricks Academy:** AI Agent Fundamentals · Databricks Fundamentals · Generative AI Fundamentals · Get Started with Databricks for Data Engineering (Jun 2026)
- **IELTS Band 7.5** | IELTS Official · **Associate Data Engineer in SQL** | DataCamp · **Data Engineer** | Hacktiv8 Indonesia · **Full-stack Web Developer** | Binar · **SQL (Intermediate)** | HackerRank
- **Languages:** Indonesian (native), English (fluent — IELTS 7.5)